

ACTIVITY 3 – DATA VISUALIZATION SCENARIOS

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Course/Code : DSA0604 - DATA HANDLING AND VISUALIZATION

Activity : Activity 3

Date : 20/07/26

SCENARIO 1: HOSPITAL PATIENT ADMISSION ANALYSIS

CODE:

```
library(ggplot2)
library(dplyr)

set.seed(42)

departments <- c("Cardiology", "Neurology", "Pediatrics", "Orthopedics", "Emergency")

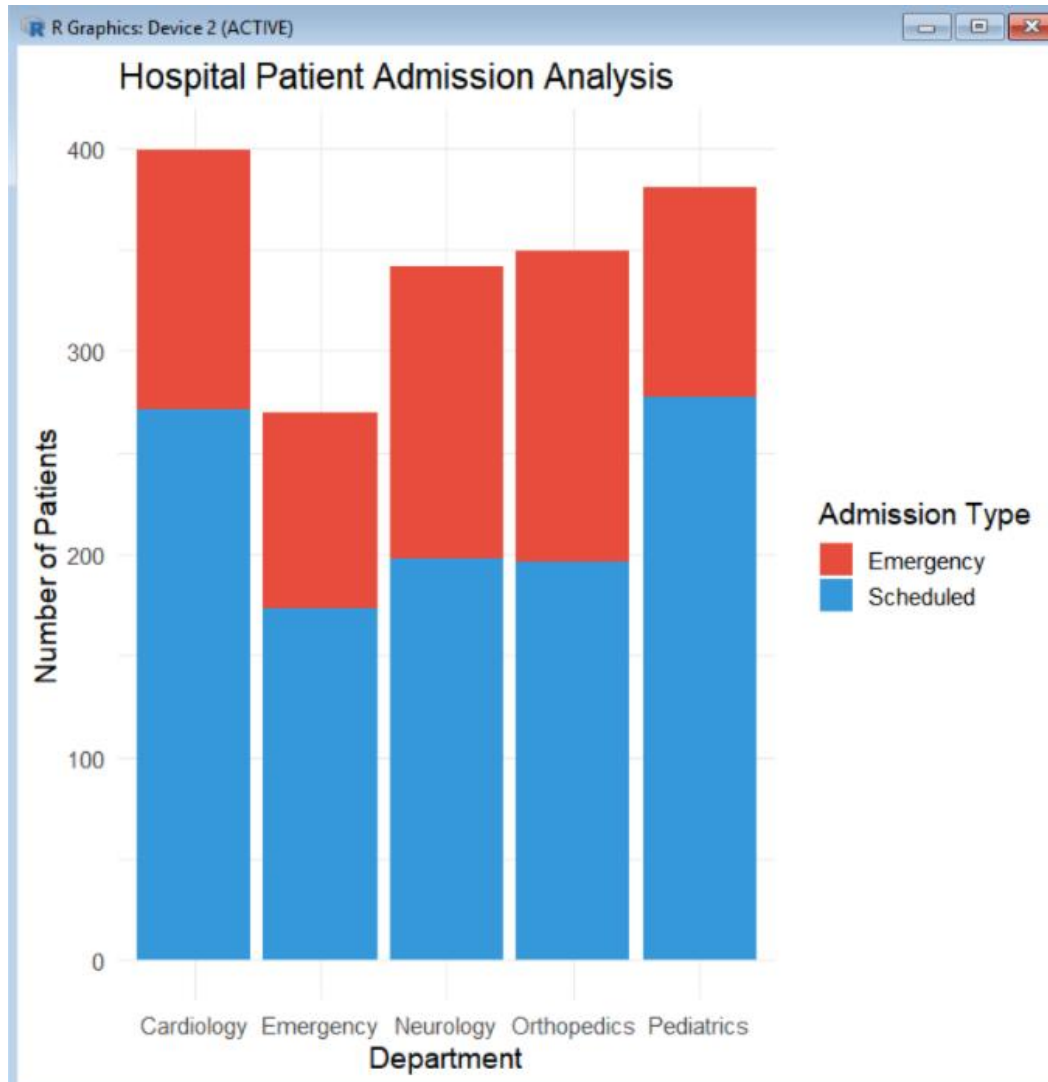
admissions <- data.frame(
  Department = departments,
  Emergency = sample(80:180, 5),
  Scheduled = sample(150:300, 5)
)

admissions_long <- admissions %>%
  tidyr::pivot_longer(
    cols = c(Emergency, Scheduled),
    names_to = "Admission_Type",
    values_to = "Patients"
  )

ggplot(admissions_long,
  aes(x = Department,
    y = Patients,
    fill = Admission_Type)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Hospital Patient Admission Analysis",
    x = "Department",
    y = "Number of Patients",
    fill = "Admission Type"
  ) +
  scale_fill_manual(values = c("#E74C3C", "#3498DB")) +
  theme_minimal(base_size = 14)
```

```
ggsave(  
  "Scenario1_Hospital_Admission_Analysis.png",  
  width = 9,  
  height = 6,  
  dpi = 300  
)
```

OUTPUT:



SCENARIO 2: E-COMMERCE CUSTOMER BEHAVIOUR

CODE:

```
library(ggplot2)
library(dplyr)

set.seed(42)
n <- 300

customer_data <- data.frame(
  Product_Category = sample(
    c("Electronics", "Fashion", "Groceries", "Home & Kitchen", "Sports"),
    n,
    replace = TRUE
  ),
  Purchase_Amount = round(rnorm(n, mean = 5000, sd = 1500), 0)
)

customer_data$Purchase_Amount[customer_data$Purchase_Amount < 500] <- 500

ggplot(customer_data,
  aes(x = Product_Category,
      y = Purchase_Amount,
      fill = Product_Category)) +
  geom_violin(trim = FALSE, alpha = 0.8) +
  labs(
    title = "Customer Spending Behaviour by Product Category",
    x = "Product Category",
    y = "Purchase Amount (₹)"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")

ggsave(
  "Scenario2_Customer_Spending_Behaviour.png",
  width = 9,
  height = 6,
  dpi = 300
)
```

OUTPUT:



SCENARIO 3: UNIVERSITY EXAMINATION PERFORMANCE DASHBOARD

CODE:

```
library(ggplot2)
library(dplyr)

set.seed(42)
n <- 300

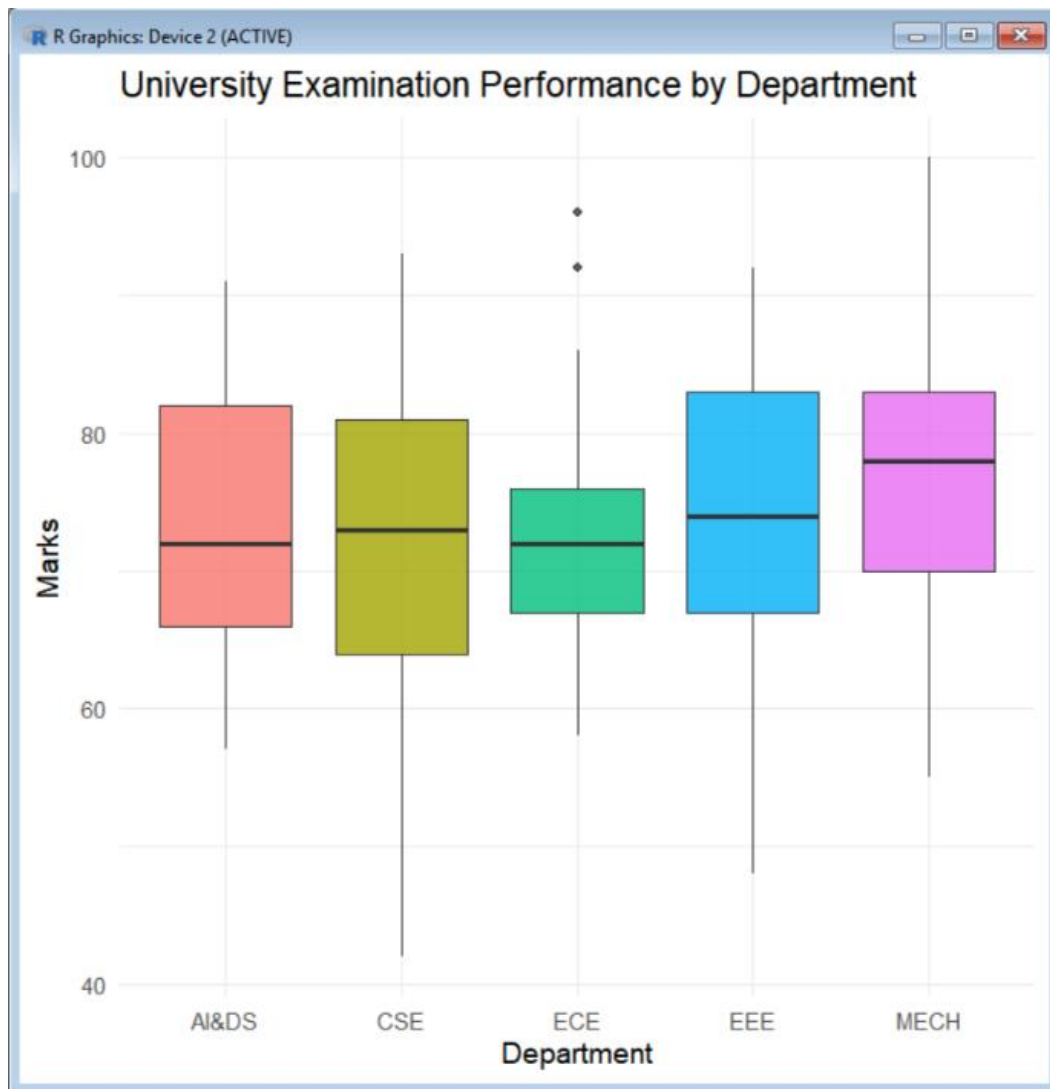
student_data <- data.frame(
  Department = sample(
    c("CSE", "AI&DS", "ECE", "EEE", "MECH"),
    n,
    replace = TRUE
  ),
  Marks = round(rnorm(n, mean = 75, sd = 10), 0)
)

student_data$Marks[student_data$Marks > 100] <- 100
student_data$Marks[student_data$Marks < 35] <- 35

ggplot(student_data,
  aes(x = Department,
      y = Marks,
      fill = Department)) +
  geom_boxplot(alpha = 0.8) +
  labs(
    title = "University Examination Performance by Department",
    x = "Department",
    y = "Marks"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")

ggsave(
  "Scenario3_University_Examination_Performance.png",
  width = 9,
  height = 6,
  dpi = 300
)
```

OUTPUT:



SCENARIO 4: CLIMATE AND RAINFALL DISTRIBUTION STUDY

CODE:

```
library(ggplot2)
library(dplyr)

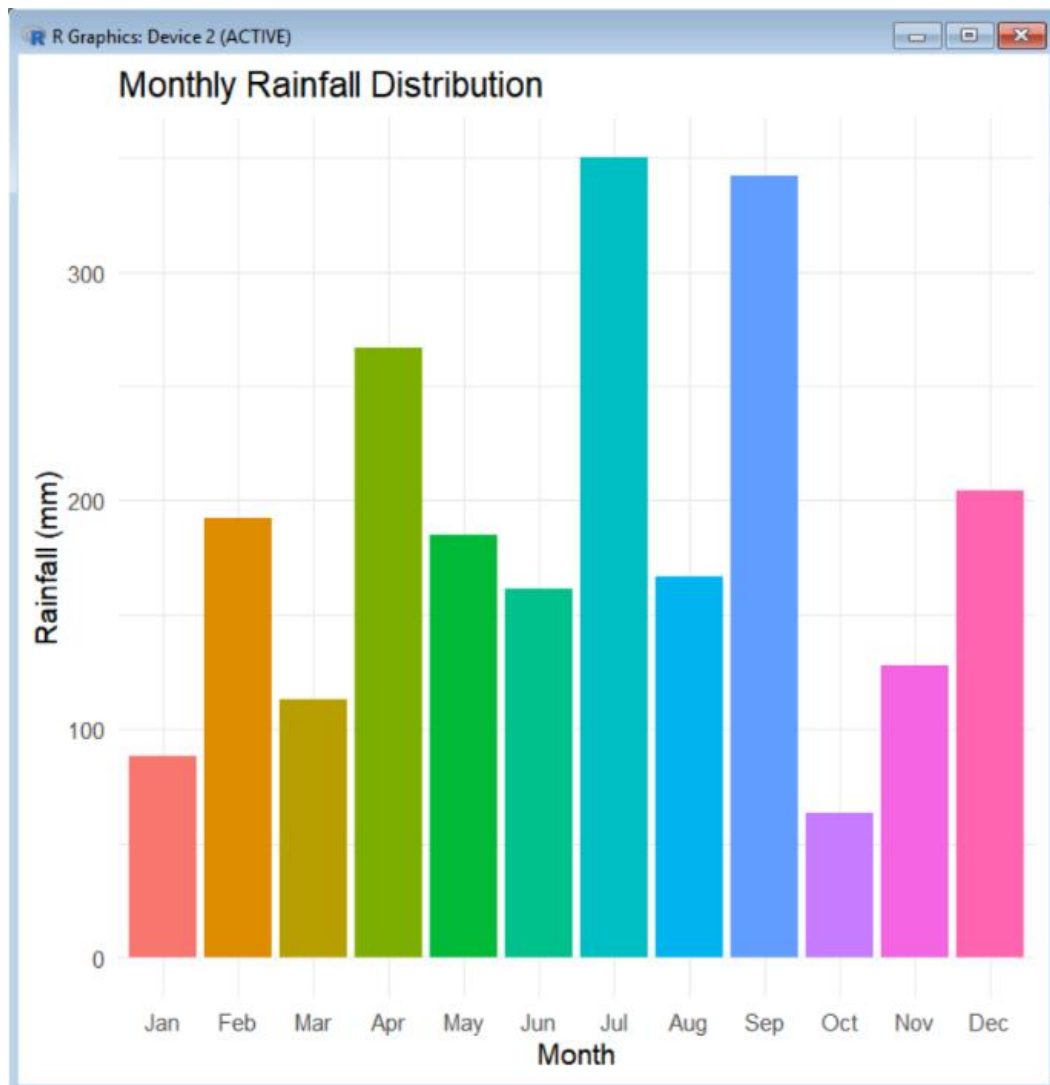
set.seed(42)

climate_data <- data.frame(
  Month = factor(
    month.abb,
    levels = month.abb
  ),
  Rainfall = sample(40:350, 12)
)

ggplot(climate_data,
       aes(x = Month,
           y = Rainfall,
           fill = Month)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Monthly Rainfall Distribution",
    x = "Month",
    y = "Rainfall (mm)"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")

ggsave(
  "Scenario4_Climate_Rainfall_Distribution.png",
  width = 9,
  height = 6,
  dpi = 300
)
```

OUTPUT:



SCENARIO 5: BANK LOAN RISK ASSESSMENT

CODE:

```
library(ggplot2)
library(dplyr)

set.seed(42)
n <- 300

loan_data <- data.frame(
  Risk_Category = sample(
    c("Low", "Medium", "High"),
    n,
    replace = TRUE
  ),
  Loan_Amount = round(rnorm(n, mean = 500000, sd = 150000), 0)
)

loan_data$Loan_Amount[loan_data$Loan_Amount < 50000] <- 50000

ggplot(loan_data,
  aes(x = Risk_Category,
      y = Loan_Amount,
      fill = Risk_Category)) +
  geom_boxplot(alpha = 0.8) +
  labs(
    title = "Loan Amount Distribution by Risk Category",
    x = "Risk Category",
    y = "Loan Amount (₹)"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")

ggsave(
  "Scenario5_Bank_Loan_Risk_Assessment.png",
  width = 9,
  height = 6,
  dpi = 300
)
```

OUTPUT:

